

Walk/Run Activity Prediction

This project predicts whether a person is walking or running using machine learning techniques. The objective is to classify human physical activities based on sensor data collected from mobile devices and wearable sensors.

Project Overview:

- Data preprocessing and Exploratory Data Analysis (EDA) performed
- Multiple machine learning models developed and evaluated
- Feature scaling and model comparison conducted for better performance

Dataset: Activity recognition dataset containing accelerometer and gyroscope sensor readings with activity labels (Walk/Run).

Tech Stack: Python, Pandas, NumPy, Scikit-learn, Matplotlib, and Seaborn.

Approach: 1. Data preprocessing and cleaning 2. Exploratory Data Analysis (EDA) 3. Feature engineering and scaling 4. Model building and training 5. Model evaluation and prediction

Results:

- Random Forest achieved the best performance
- High classification accuracy on test data
- Successful prediction of walking and running activities

Future Work:

- Hyperparameter tuning for improved accuracy
- Real-time activity monitoring system
- Deployment using Streamlit or Flask

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